

Designing the Kinetics of Biomolecular Interactions

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The function of complex biomolecules depends heavily on their conformational changes. Consequently, precise design of biomolecular function requires complete characterization and fine-tuning of their kinetic properties. Being a standard tool for studying the kinetic mechanisms of functional conformational changes, molecular dynamics simulations are typically considered incapable of molecular design due to its extremely low computational efficiency. Thanks to the advancements in path-searching (enhanced sampling) methods in recent years, this efficiency bottleneck has been largely alleviated. In this talk, I will review the development of the Travelling-salesman based Automated Path Searching (TAPS) method, demonstrate its key role in elucidating the mechanism of biomolecular signalling, the design of specific agonists for G-Protein-Coupled Receptors and the design of complex enzymes, such as high-fidelity CRISPR-Cas variants. Finally, I will also discuss the main challenges ahead of kinetics design and its foundation value for bio-AI of the next-generation.